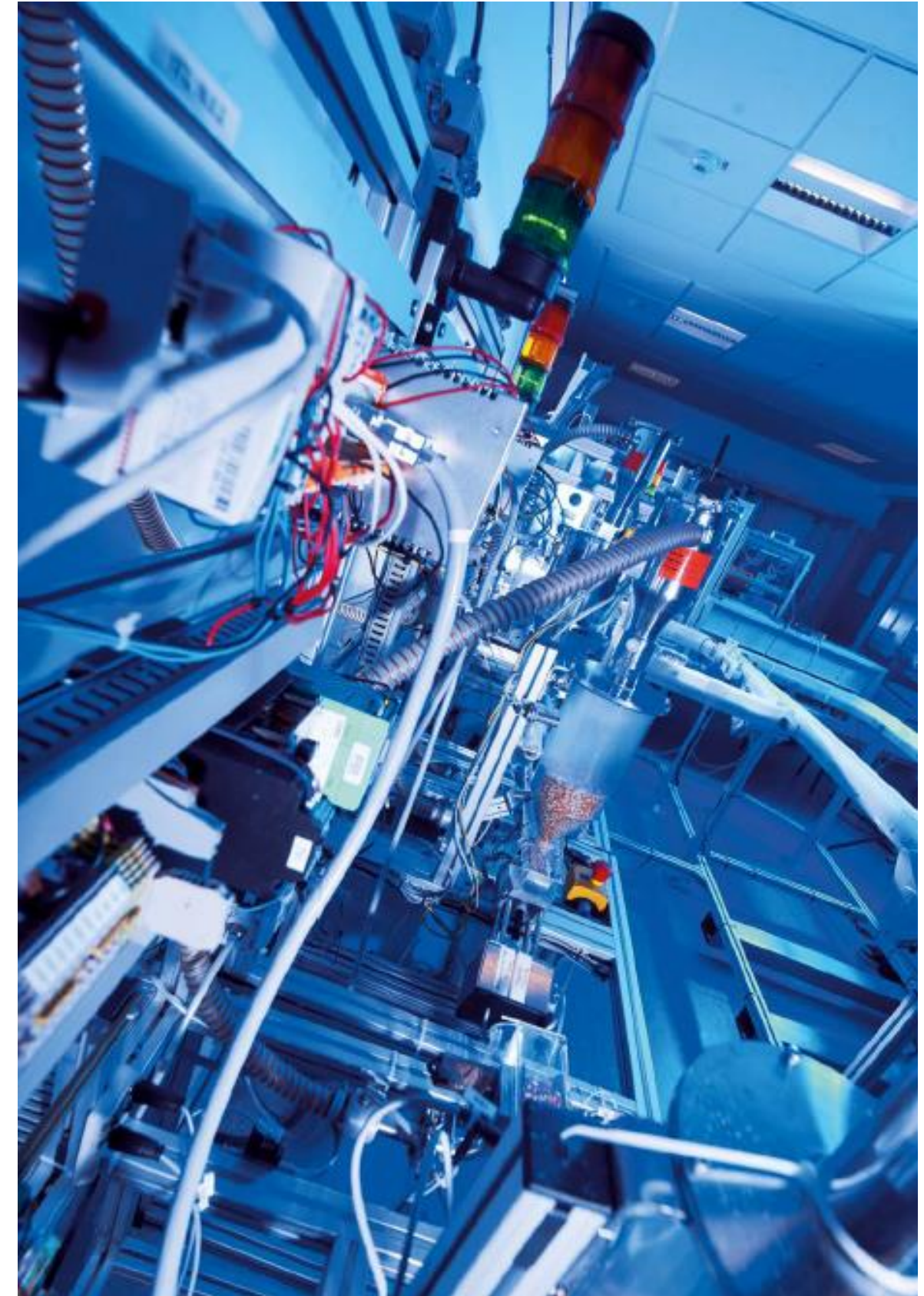


Echtzeit-Datenverarbeitung

Cyber-Security

Prof. Dr. Henning Trsek

Technische Hochschule Ostwestfalen-Lippe
Fachbereich Elektrotechnik und Technische Informatik
inIT - Institut für industrielle Informationstechnik
henning.trsek@th-owl.de





Security
Einführung

Security in
der OT

Sichere SW
Entwicklung

Agenda

Security Einführung

Warum Security in isolierten industriellen Systemen?

Schwachstellen und Bedrohungen in industriellen Systemen

Gemeinsamkeiten und Unterschiede

Best practices und Standards

- ▶ IEC 62443
- ▶ VDI 2182 – Vorgehensmodell
- ▶ Fallstudie zur VDI 2182

Definition Information Security

Information security

Ensures that the security objectives **Confidentiality**, **Integrity** and **Availability** of all information assets of an organization are always met.

Information assets are

Information and information processing systems, which have a certain **value** for the organization

Also business processes are concerned, as long as they depend on the previous systems and information.

Information in **all possible types and formats** are concerned: printed, spoken, audio, video, IT-Systems

Security objective: Confidentiality

Definition

- ▶ Information assets are only available and disclosed to authorized persons and systems

Considered in terms of

- ▶ Transport, storage, processing

Typical control

- ▶ Encryption

Objective violated, if information assets are disclosed to unauthorized persons/systems

Security objective: Integrity

Definition

- ▶ Information assets are always complete and correct, an unnoticed modification by unauthorized persons and systems is impossible

Considered in terms of

- ▶ Transport, storage, processing

Typical control

- ▶ Check sums

Objective violated, if information assets are modified by unauthorized persons/systems

Security objective: Availability

Definition

- ▶ Information assets can be accessed and used by authorized persons and systems as intended

Considered in terms of

- ▶ Data, services, infrastructures, etc.

Typical control

- ▶ Redundancy, over provisioning

Objective violated, if information assets are only limited or not at all useable as intended due to an attacker

Agenda

Security Einführung

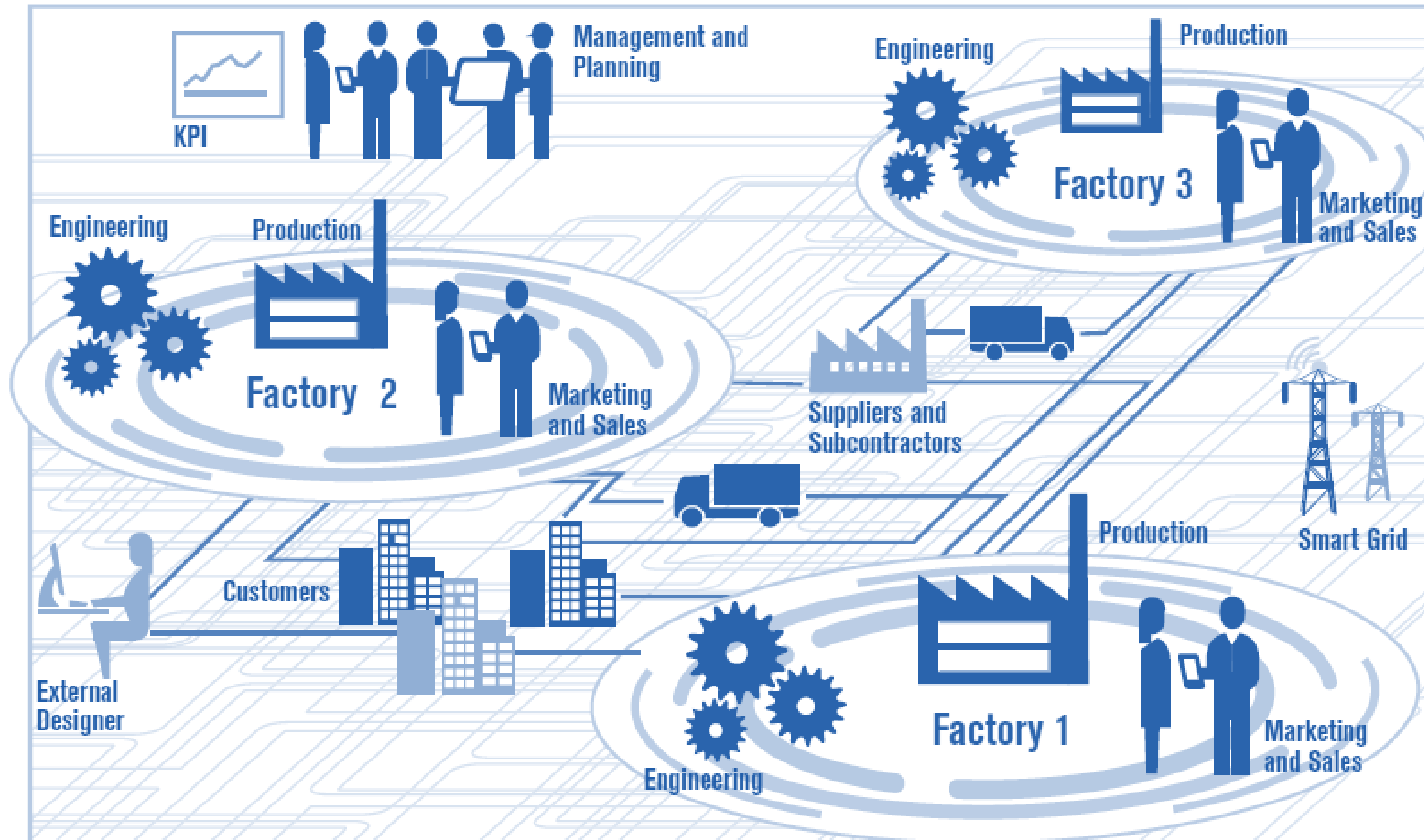
Warum Security in isolierten industriellen Systemen?

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Industry 4.0 bedeutet Vernetzung aller Komponenten



Source: Hewlett Packard 2013

- ▶ Alles ist miteinander verbunden
- ▶ Vertikale und horizontale Integration
- ▶ Smarte und vernetzte Produkte
- ▶ Verfügbare Daten werden zum Monitoring der Produktion verwendet

Die aktuelle Situation in der OT

IMMER MEHR CYBER
IT-Secur
Im Zeitalter der
Cyberattacken.

Flugzeugbau-Zulieferer stoppt Produktion – Verdacht auf Ransomware

Die belgische Firma ASCO Industries leidet seit Freitag unter einer nicht näher spezifizierten Attacke, die ihre Produktion zum Erliegen gebracht hat.

Lesezeit: 1 Min. In Pocket speichern



(Bild: ASCO Industries)

UPDATE 13.06.2019 15:39 Uhr

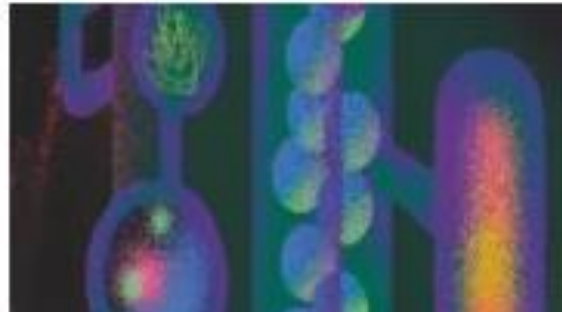
Hand interacting with a laptop displaying a dashboard with charts and graphs, symbolizing industrial cyber security.

heise +

Triton: Malware

Der Schadcode Triton kann Unfälle verursachen. Er ist eine Bedrohung.

Lesezeit: 4 Min. In Pocket speichern



MB Connect Line

The most recent company known to have their software infected with the Havex backdoor was the German company [MB Connect Line GmbH](#), who are known for their industrial router mbNET and VPN service mbCONNECT24.

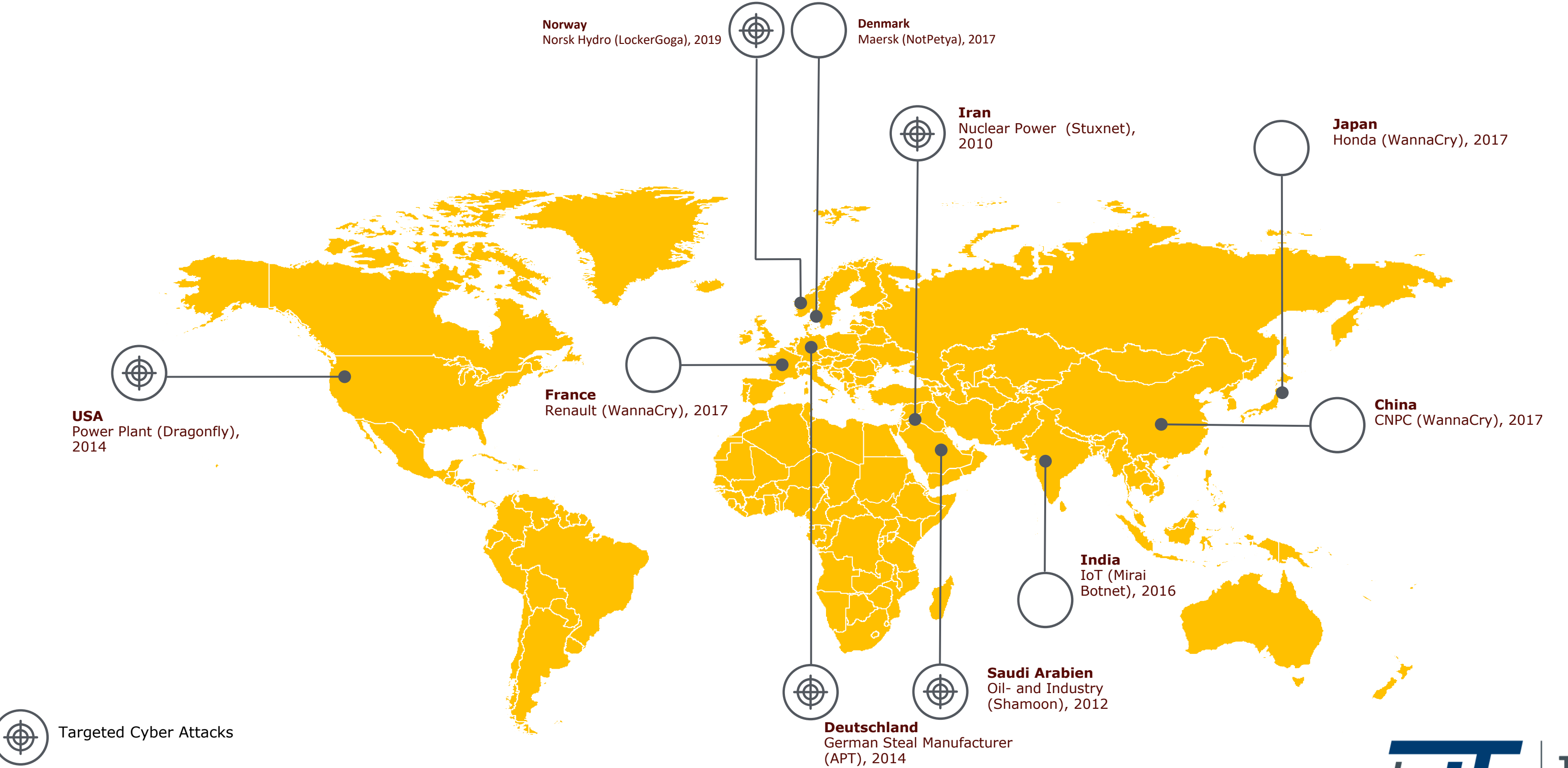
MB Connect Line published a [report](#) about the Dragonfly intrusion in September 2014, where they write:

On 16th of April 2014 our website [www.mbconnectline.com](#) has been attacked by hackers. The files mbCHECK (Europe), VCOM_LAN2 and mbCONFTOOL have been replaced with infected files. These files were available from 16th of April 2014 to 23th of April 2014 for download from our website. All of these files were infected with the known Trojan Virus Havex Rat.



Image: Screenshot of trojanized mbCONFTOOL installer from our sandbox execution

Globale Cyber Angriffe auf ICS Infrastrukturen



 Targeted Cyber Attacks

Haben Sie schon von Shodan gehört?

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Explore the Internet of Things

Use Shodan to discover which of your devices are connected to the Internet, where they are located and who is using them.



See the Big Picture

Websites are just one part of the Internet. There are power plants, Smart TVs, refrigerators and much more that can be found with Shodan!



Monitor Network Security

Keep track of all the computers on your network that are directly accessible from the Internet. Shodan lets you understand your digital footprint.



Get a Competitive Advantage

Who is using your product? Where are they located? Use Shodan to perform empirical market intelligence.



56% of Fortune 100



1,000+ Universities

Shodan is used around the world by researchers, security professionals, large enterprises, CERTs and everybody in between.

Zusammenfassung

- ▶ Weltweite Angriffe auf ICS sind die Realität
- ▶ Wenn eine Verbindung zum Internet besteht, werden die Systeme gefunden
- ▶ Kleine und mittlere Unternehmen sind zunehmend im Fokus
- ▶ Standard Protokolle und sehr spezifische Protokolle werden angegriffen
- ▶ Unternehmen werden ohne Vorauswahl angegriffen
- ▶ Unternehmen werden observiert und zu einem passenden Zeitpunkt angegriffen

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BSI Top Ten Bedrohungen

Critical threats	Scenarios
Malware Infection via Internet and Intranet	Exploitation of known weaknesses, manipulation of external web pages, untargeted malware
Introduction of Malware on Removable Media and External Hardware	Infected USBs, maintenance computers are infected
Social Engineering	Phishing, spear-phishing, unauthorised access
Human error and sabotage	Incorrect configuration, uncoordinated patching
Intrusion via remote access	Direct attack, indirect attack
Control components connected to the Internet	Retrieval of control components, exploitation of known weaknesses
Technical malfunctions	Defects of components, software errors
Compromised Smartphones	Theft or loss, attack on the Smartphone
Compromised Extranet or Cloud	Disruption of communication, insufficient external security mechanisms
DDoS attacks	Individual components of the systems

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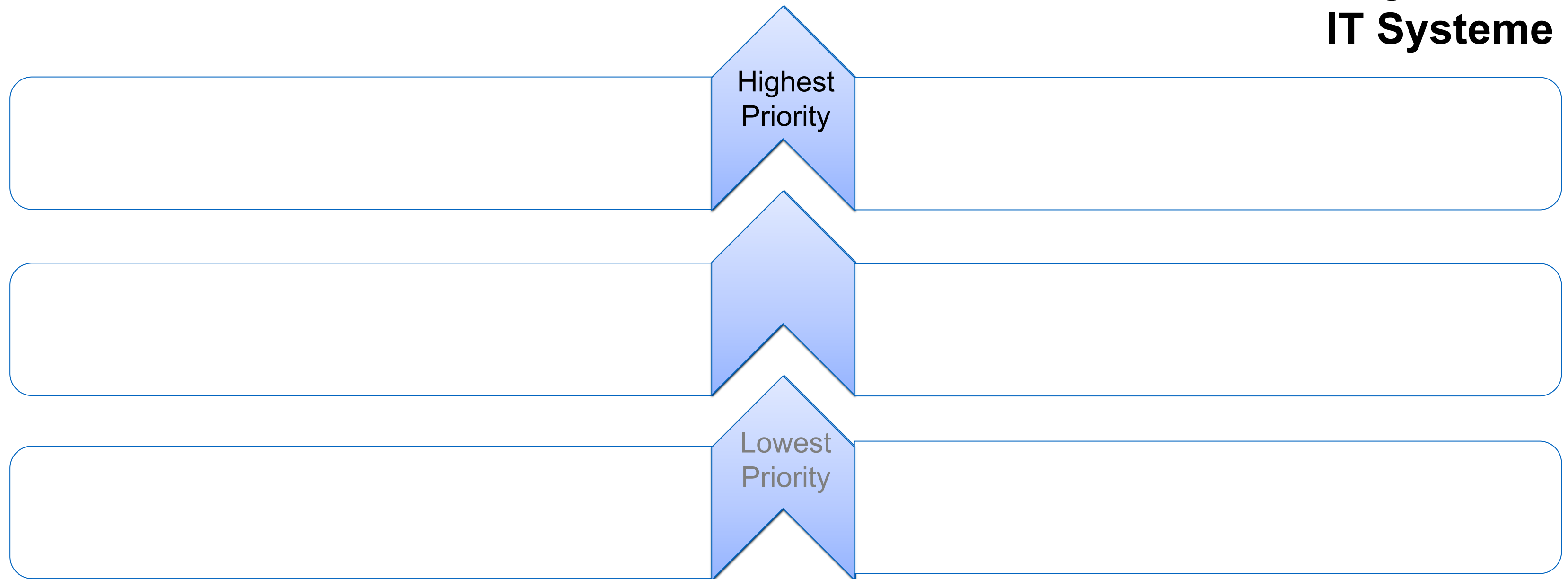
Gemeinsamkeiten und Unterschiede

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Unterschiede - Security Schutzziele

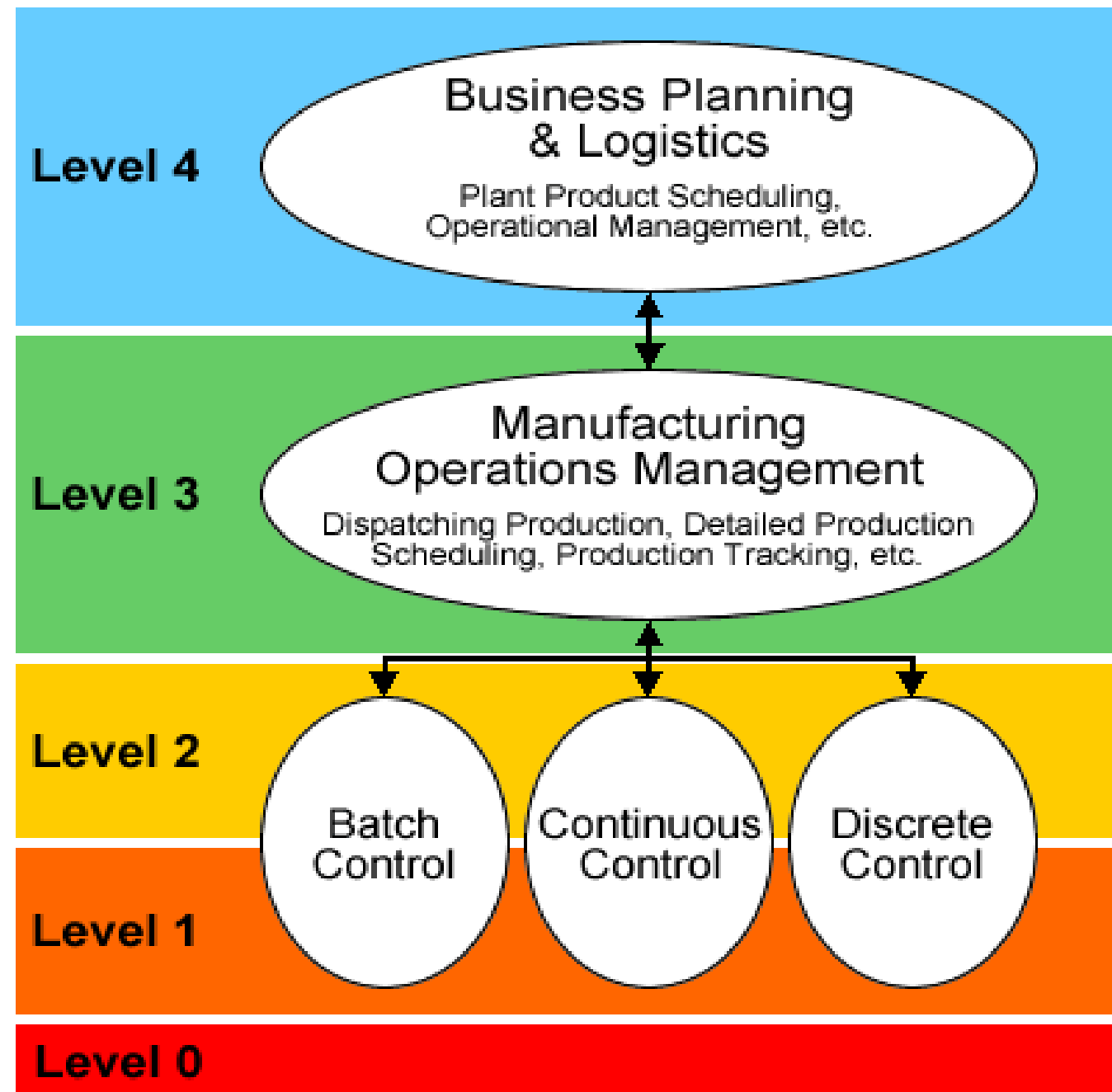
Industrielle Automation

**Allgemeine
IT Systeme**



Aktuelle Situation und Verantwortlichkeiten?

Enterprise Integration ISA 95 / IEC 62264



Typische Verantwortlichkeiten

► Enterprise IT

- IT security für alle IT Systeme innerhalb des Unternehmens

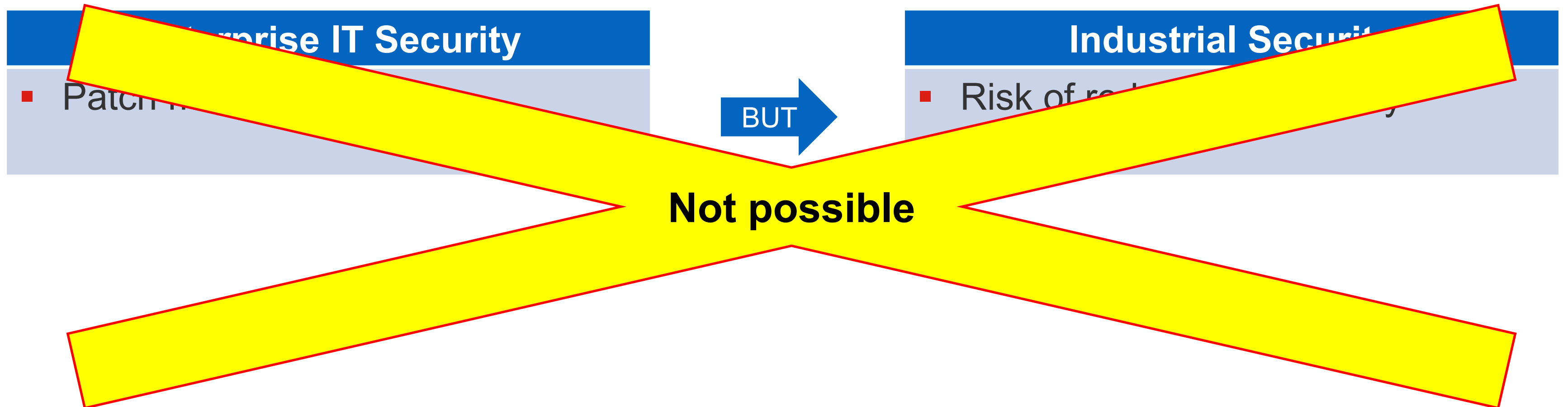
► Production

- Betrieb der Fertigung inkl.
 - DCS/MES Betrieb
 - “Automation Security”
 - Availability
 - Safety

Wir können doch bestehende Security Maßnahmen nutzen...

Standard controls of IT security applied to manufacturing

Example: Protection against Malicious Code



IT Security und Industrial Security – Ähnlich aber nicht gleich

Similarities

Office IT Systems

Industrial Control Systems (OT)

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7 easy steps to ICS security

Step 1

- Assess existing system

Step 2

- Document policies and procedures

Step 3

- Train personnel and contractors

Step 4

- Segment the control network

Step 5

- Control access to the system

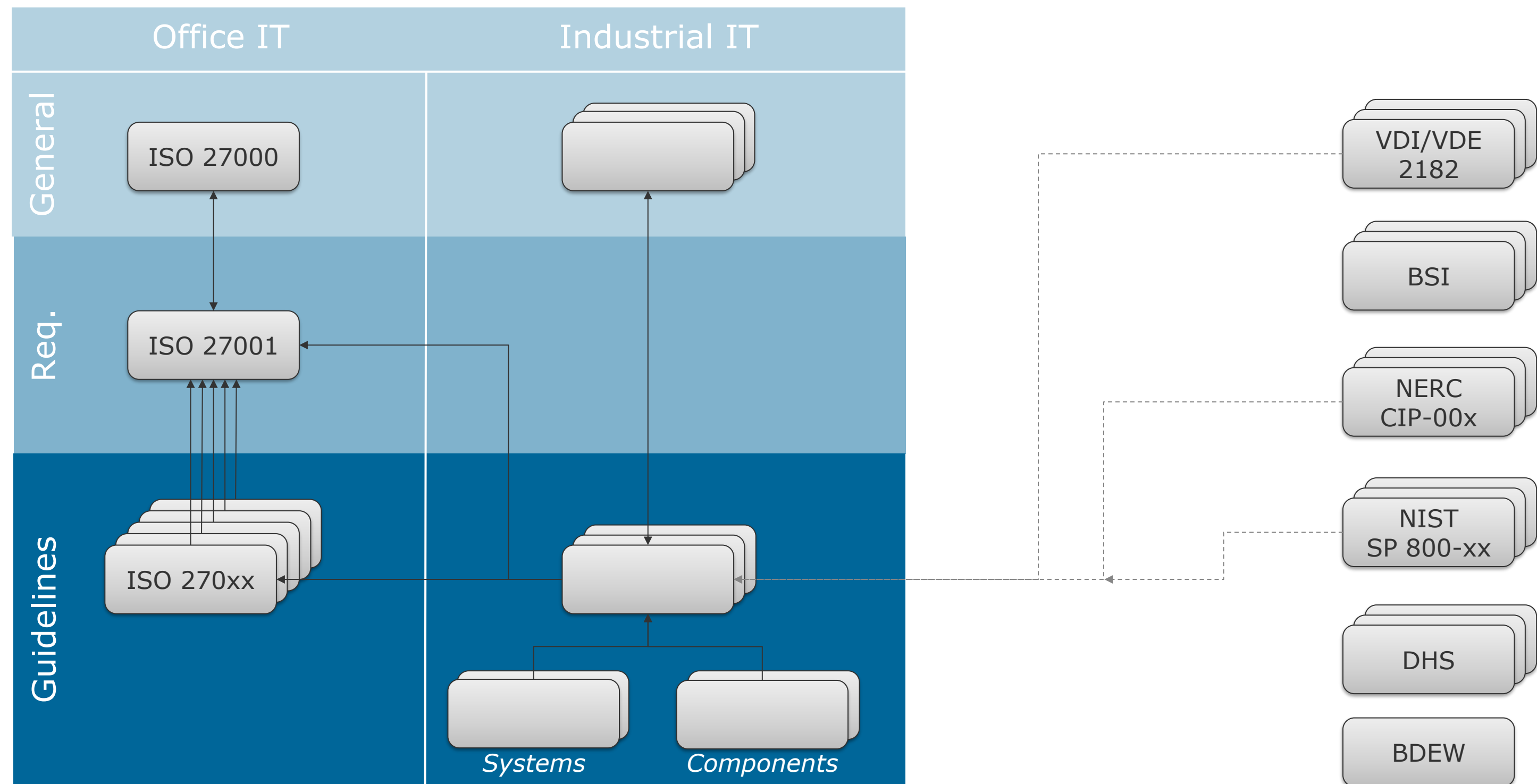
Step 6

- Harden the components

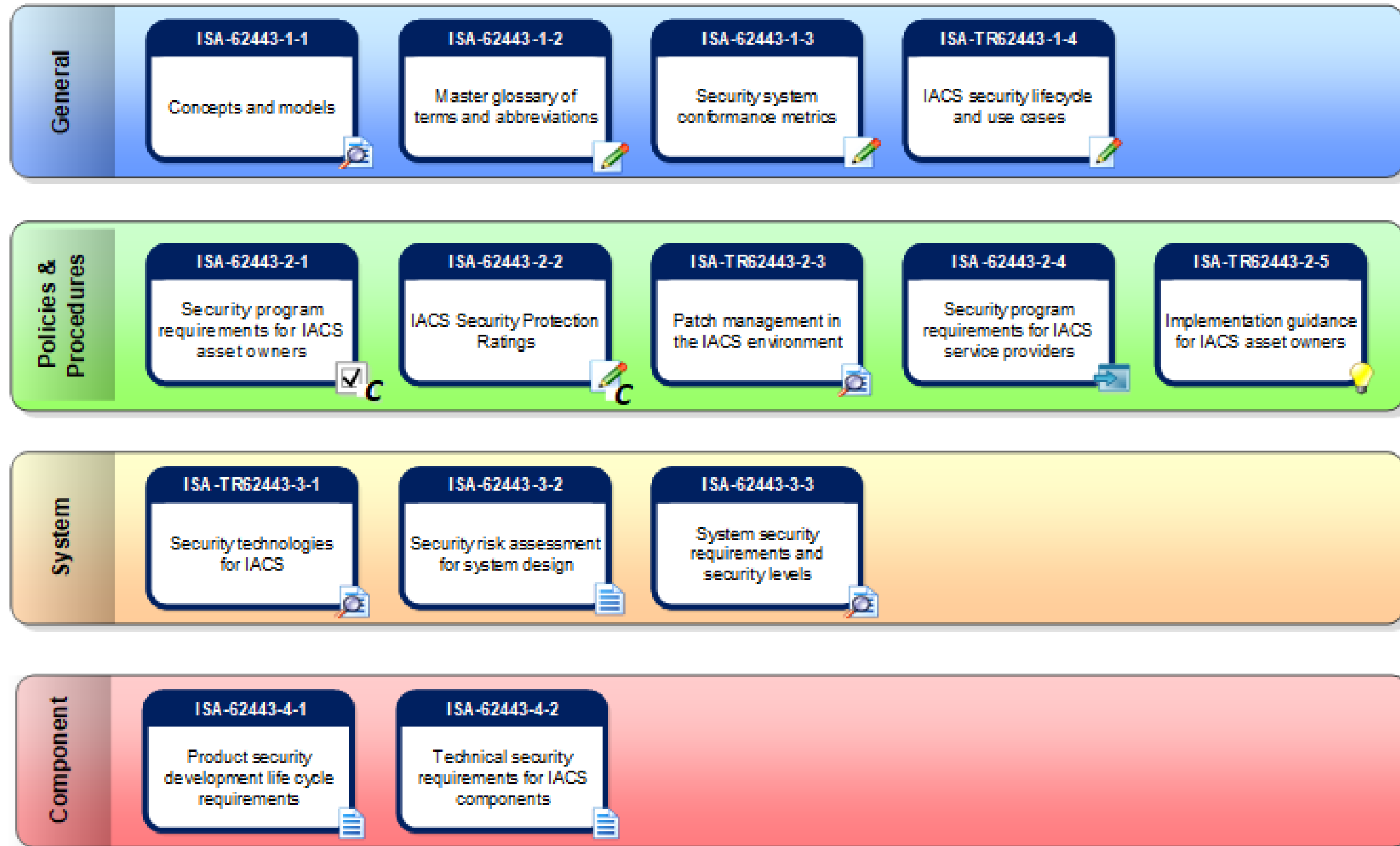
Step 7

- Monitor and maintain security

Aktivitäten in der Standardisierung



Standardization IEC 62443



Source: ISA 99, IEC 62443.

Industrial Security - Zusammenfassung

- ▶ Security ist integraler Bestandteil von Industrie 4.0
- ▶ Industrie 4.0 bedeutet vernetzte und offene Systeme
- ▶ Security ist niemals statisch
- ▶ Cyber security management system (CSMS) ist ein wichtiger Schritt
- ▶ Security awareness